

SPECIFICATIONS: Mechanical and Electrical instruments

Scope of work

Supply of “**Mechanical and Electrical instruments**” as per specifications mentioned below.

SPECIFICATIONS FOR MECHANICAL AND ELECTRICAL INSTRUMENTS

The instruments provided should be Remi, Cumi, Thermofisher or Equivalent make. Given below is the list of standard instruments along with their specifications and quantity.

Firms shall quote individual price for all line items. Price will be evaluated on total lot wise basis.

S.No	Description	Qty.
1	Ultra Sonic cleaner Capacity:6.5L (min.) Tank Size:12'' x 6'' x 6'' (min.) Bath Temperature upto 90 degree and beyond Step less power controller Power Supply 170VAC-270VAC Frequency Range of 18 KHz. to 68 KHz.	1
2	Vacuum Dessicator Material: Polypropylene/ Polycarbonate Capacity: 300 mm (min.) Silicone O Rings to work upto temperature of 0 degree	1
3	Mechanical Stirrer (High Torque) Stirrer with 25 LTRS Stirring Capacity and PMDC Motor Stirring Fluid: - Colloidal and Resins, Stirling Shaft Ø x Length(mm): - 10X550 Stirring Capacity (Ltrs.): - 25 Impeller: - RADIAL TURBINE Min. Speed (RPM) :- 50 & Max. Speed (RPM) :- 550	2
4	AC Stirrers Stirling shaft dia* Length(mm): 10X730; Stirring capacity (Ltrs.): 40 (min) Impeller: Radial turbine; Motor: A/C Brushless Min. speed (RPM): 100; Max. speed (RPM): 1400 or higher; Digital speed indicator: Yes	2
5	Magnetic Stirrers (With Hotplate)	3

	Handling Capacity: 500ml to 2 Ltrs Min Voltage: 220 - 240 V Power: 150 - 500 W Operating Frequency: 50 Hz Max Speed: 1200 rpm or higher Stirring capacity: 5 Ltrs or higher Heating capacity: min 500 watt	
6	Rotary Barrel Pump Operation: Dual Direction Source: Manual Style: Centrifugal Discharge Rate: 4L/20 turns (min.) Outlet - 3/4 Inch Inlet - 1 Inch	2
7	Plastic Lever Pump (for viscous fluids) Discharge/ Flow rate: 500cc/ stroke (min) Material: PVC/ Polypropylene Suitability: High Viscous resins and oils Hose size: 3/4 inch x 2 m	2
8	Welding Face Shield Clear Visor Film with Anti Splash Guard Face Cover Application: Face Safety Type: Respiratory, Safety, Foldable, Disposable Nature: Reusable Dimension: 6" X 9", 9" X 12"	3
9	Wrap Roll EP EPE Foam Wrap Roll 5MM Thickness 2 Feet X 20 Feet Set (white colour)	10
10	Magnetic Stirrer with glass ceramic hot plate Speed Range (rpm): 1500 or higher Hotplate Surface: glass ceramic Heating power (W) :1000 or higher Max. Temperature Range: 550 °C or higher	3
11	Magnetic beads For magnetic stirrer with glass ceramic plate Temperature range: - 20°C to 550°C. Size: -20mm, 30 mm, 50mm, 60 mm and 76mm	3 nos. each size
12	Orbital Sanders	2

	Grit Type : Fine Power Source : Air-Powered Maximum Rotational Speed: 10000 rpm Disc Diameter: 125 mm (max.) Hook and Lock back up pad	
13	Infrared Thermometer Temperature range: -30°C to 350 °C Response time < 500ms Spectral response: 8mm to 14mm Emissivity: 0.10 to 1.00 Display resolution: 0.1 °C Repeatability: +/- 1.0% Accuracy: 0.01 °C Usage: Industrial Battery Life: Min. 12 hrs Operating temperature: 0- 50 degree Celsius Operating Humidity: 10% to 90% Rh Laser Safety: FDA and EN 60825-1 Class-II Ingress Protection rating IP 40 per IEC 60529	2
14	Precision Air blowgun Size: 4 inch Type: Pneumatic Pressure: 0- 8 Kg	3
15	Air flow nozzle connector for pipe systems Material type: Stainless Steel/ Polycarbonate 5 different sizes: a) Inlet 4mm to outlet 6mm (Type T, Type Y & Type I connection) b) Inlet 6mm to outlet 8mm (Type T, Type Y & Type I connection) c) Inlet 8mm to outlet 10mm (Type T, Type Y & Type I connection) d) Inlet 10mm to outlet 12mm (Type T, Type Y & Type I connection) e) Inlet 12mm to outlet 14mm (Type T, Type Y & Type I connection) Each size contain 3 different type making it a set. Pressure capacity: min. 8 bar Total: 5 set per each size (Total 75 nos.)	5 set (per each size) (Total- 25 sets)
16	C-Clamp 10/250 heavy duty Dimension: 40 x 16 x 3 centimeter Material: Carbon steel (Drop forged) Size: 250 mm Tensile strength: 1500 Pounds	5
17	Moptar Pestle 6"	2

	Material: High strength ceramic Finish: Glaze	
18	Digital pH meter (Electrode Type) Voltage: 220V – 230V Type: Table-top pH range: 0 – 14 Display type: LCD Calibration: 2 points Resolution: 0.1 pH 0.1 mV Temp. 0.1°C Accuracy: +0.1 (min) Application: laboratory Additional accessories: pH Electrodes with stand, dust cover	2
19	Soldering station Output Power: 650 W Power rating: 1200 Watt Temperature range: 100°C to 480°C Temperature stability: $\pm 1^{\circ}\text{C}$ Display type: LED Soldering stand: SS Rated voltage: AC 220V $\pm 10\%$ 50 Hz Output power: 550 Watt (min) Heating element: Ceramic Pre-heating platform: 120 x 120 mm Temperature sensor type: K type	1
20	Funnel plastic for oil Design type: Flexible (Bendable) Material: Plastic Ergonomic handling, anti-splash & wide bowl opening design	5
21	Bearing puller set Material: Carbon Steel Segment grip type Collectors Collect Size:- 8-11mm, 12-17mm, 18-23mm, 24-29mm, 30-34mm, 34-38mm, 39-43mm, 44-48mm, 49-53mm, 54-58mm. 3 pcs screw adaptors Thread Adapters: M10*1.5p, M8*1.5p, M6*1.5p Should include: 1 Screw arbor, 1 slide hammer, 1 debug rod, 2.5 Jaws Assembly, 2 jaw puller(L-leg puller), Bar-type bearing splitters, hex push-puller legs	1 no.

Table1: Mechanical and Electrical instruments

1. **Delivery Terms:** Within 120 days from receipt of SO at CEMC building no. 0603, C/o RCI Campus, Hyderabad.
2. Firm should deliver the items in Single Lot.
3. **Payment Terms:** DRDO / ASL standard payment terms applicable.
4. **Inspection:** Offered tools should be a standard one which shall be inspected at ASL.
5. **Warranty:** As provided by OEMs.
6. **Undertaking:** A self certificate that the drawn specifications are not tailored to suit a particular brand and is generic to attract competitive bidding if the bidding mode is LBM and OBM.